Pitta Jaya Prakash

+91 9603589705 | pittajayaprakash9@gmail.com | LinkedIn | GitHub | Hyderabad, Telangana, India

Professional Summary

DevOps Engineer with SRE experience and 4+ years in cloud infrastructure, CI/CD automation, and Infrastructure as Code (IaC). Proven track record of reducing deployment times by 80%, achieving 99.9% system uptime, and optimizing cloud costs by 20%. Hands-on expertise with AWS, Kubernetes, Docker, Terraform, Jenkins, Prometheus, and Grafana across high-availability microservices architectures.

Technical Skills

Cloud Platforms: AWS (EC2, S3, IAM, VPC, Lambda, RDS, EKS, CloudFormation, CloudWatch, Route 53, ALB), GCP
CI/CD Tools: Jenkins, GitLab CI, GitHub Actions
Infrastructure as Code: Terraform, AWS CloudFormation
Containers & Orchestration: Docker, Kubernetes, AWS EKS, Helm, Horizontal Pod Autoscaler (HPA), Cluster Autoscaler
Monitoring, Observability & Logging: Prometheus, Grafana, Alertmanager, PagerDuty, Splunk, AWS CloudWatch, ELK Stack (Elasticsearch, Logstash, Kibana), Distributed Tracing
Scripting Languages: Python, Bash, Shell Scripting
Version Control: Git, GitHub, GitLab, Bitbucket
Operating Systems: Linux (CentOS, Ubuntu, RHEL), Windows, macOS
Networking & Security: VPC Design, IAM Security, SSL/TLS, DNS (Route 53), Load Balancing, Network Policies
Other: Remote State Management (S3 + DynamoDB), Microservices Architecture, High Availability (HA), Disaster Recovery (DR)

Work Experience

DevOps Engineer (with SRE Exposure) — Feb 2022 – Present
Tata Consultancy Services (TCS) — *Hyderabad, India*

- Automated CI/CD pipelines using **Jenkins and GitHub Actions**, reducing deployment time by **80%** and improving release frequency from monthly to daily deployments.
- Orchestrated containerized applications using **Docker and Kubernetes (EKS)**, enhancing scalability and deployment consistency across 15+ microservices with zero-downtime rolling deployments.
- Implemented **Infrastructure as Code (IaC)** with Terraform, automating provisioning and configuration management for **50+ AWS resources**, reducing provisioning time from hours to minutes.
- Monitored system performance using **Prometheus and Grafana**; supported SRE practices including SLO/SLI tracking and incident response, achieving **99.9% uptime** and reducing MTTR by 40%.
- Secured AWS environments by applying best practices in **IAM, VPC, and S3 bucket policies**, reducing security vulnerabilities by 30%.
- Integrated **ELK Stack and Splunk** for centralized log aggregation, enabling faster root cause identification across distributed microservices.
- Collaborated with development teams on **capacity planning** and right-sizing of AWS resources, achieving **20% reduction in cloud costs**.

Key Projects

CI/CD Pipeline Automation for Microservices | *Jenkins, Docker, Kubernetes, GitHub Actions, Git*

- Designed and implemented multi-stage Jenkins pipelines for automated builds, security scanning, testing, and deployments across dev, staging, and production environments.
- Integrated Docker containerization with Kubernetes rolling deployments and automated rollback on health check failures.
- Established Git-based triggers and branch protection strategies for automated pipeline execution, reducing manual intervention by 90%.
- **Result:** Reduced deployment time by 80%, improved release frequency to daily, and improved rollback efficiency across all environments.

Kubernetes Cluster with Autoscaling & High Availability | *Kubernetes, AWS EKS, Prometheus, Grafana*

- Deployed and managed production-grade Kubernetes clusters on AWS EKS with multi-AZ node groups for high availability.
- Implemented Horizontal Pod Autoscaler (HPA) and Cluster Autoscaler for dynamic scaling based on workload metrics.
- Configured liveness and readiness probes, pod disruption budgets, and resource quotas to ensure application resilience.
- **Result:** Achieved 40% better resource utilization, zero-downtime deployments, and seamless scaling during peak traffic.

Infrastructure as Code with Terraform | *Terraform, AWS, Git, S3, DynamoDB*

- Developed modular, reusable Terraform modules for AWS services including EC2, S3, VPC, IAM, RDS, and EKS.
- Implemented remote state management with S3 backend and DynamoDB state locking for safe team collaboration.
- Integrated Terraform workflows into CI/CD pipelines with automated plan and apply stages and drift detection.
- **Result:** Reduced infrastructure provisioning time from hours to minutes with fully reproducible, version-controlled environments.

Education

Kakinada Institute of Engineering and Technology — Kakinada, Andhra Pradesh
Master of Technology (M.Tech) – CGPA: 8.0/10 — *Graduated: Dec 2024*

Kakinada Institute of Engineering and Technology — Kakinada, Andhra Pradesh
Bachelor of Technology (B.Tech) – CGPA: 7.89/10 — *Graduated: Jun 2021*

Certifications

AWS Certified Solutions Architect – Associate | Amazon Web Services

AWS Certified Cloud Practitioner | Amazon Web Services